

Temporal Patterns and Pixel Precision: Satellite-Based Crop Classification Using Deep Learning and Machine Learning

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Declarations

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Abstract

Accurate crop classification from satellite imagery is essential for agricultural monitoring, yield forecasting, and food security assessment, yet remains particularly challenging in dryland farming regions of the Global South where spectral similarity between crop types, irregular phenological cycles, and persistent cloud cover complicate classification. Most crop-classification studies report performance under random or field-wise cross-validation *within* a single region, which measures in-distribution fit rather than the spatial transfer that operational mapping actually requires. Using a labeled field-boundary dataset from the Western Cape of South Africa, we benchmark nineteen classical, deep learning, and patch-based models twice: under conventional in-region field-wise validation, and on a spatially disjoint holdout tile. The two protocols produce sharply different conclusions. Under in-region validation the dense temporal deep nets rank at the top (macro-F1 0.72–0.78); on the spatial holdout they collapse, and the CNN-BiLSTM ensemble — the model conventional field-level evaluation singles out as best — suffers the largest degradation of any model (macro-F1 0.72 to 0.46, a 0.26 drop), falling to near-worst. The ranking inverts: a TabNet ensemble becomes the best single model (macro-F1 0.60, Kappa 0.54), and simple classical baselines — gradient-boosted trees and even logistic regression (macro-F1 0.56) — transfer with the smallest gaps. We argue this is not incidental but mechanistic: the sparse, axis-aligned feature selection that made Random Forest and gradient-boosted trees the remote-sensing workhorse for two decades is precisely the inductive bias that governs spatial transfer, and TabNet inherits it (attentive sparsemax feature masks) while dense CNN/RNN/attention temporal nets discard it and overfit region-specific signal. A complementary lever — field-level aggregation as variance reduction — recovers transfer for dense models (L-TAE improves out-of-sample when aggregated) but adds nothing for already-sparse models. We further show, via a training-set reduction experiment, that the most transferable models retain near-full holdout accuracy with as little as 25% of the training fields. Classical models — including XGBoost, Random Forest, and LightGBM — were paired with automated time-series feature extraction via `xr_fresh`, evaluated here for the first time in crop classification. The central message for the remote-sensing community is that model selection in crop mapping is an artifact of the validation protocol: spatial generalization, not in-distribution accuracy, must drive it.

Keywords: Crop classification, Satellite imagery, Deep learning, Machine learning, Remote sensing

1 Introduction

Accurate and timely crop classification is foundational to precision agriculture, enabling yield forecasting, resource optimization, food security assessment, and climate-resilient decision-making [1, 2]. The widespread availability of high-resolution Earth observation data, especially from the Sentinel-2 satellite constellation, has transformed the landscape of agricultural monitoring. With 13 spectral bands, spatial resolutions ranging from 10 to 60 meters, and a revisit frequency of approximately

five days, Sentinel-2 offers dense multi-temporal imagery rich in spatial and spectral information.

Despite these advancements, reliable crop classification remains a complex task on plots throughout Africa and the Global South. Spectral similarities between crops, temporal variability in planting and phenological cycles, and atmospheric disturbances such as cloud cover hinder classification accuracy. In Africa, this is compounded by the complexities of rain-fed dryland farming [52]. Traditional machine learning methods often rely heavily on manually engineered features, such as vegetation indices or statistical summaries, which may capture pertinent temporal and spectral dynamics but are often more limited in complexity of features. A number of methods like XGBoost (XGB) and Random Forest (RF) have proven effective in a variety of crop mask and classification tasks [39, 40, 41]. However in regions with significant differences in cropping patterns and practices, the complexity of the task can outstrip the exploitable variability in the training data [42, 39]. Newly released automated feature extraction methods like that implemented in `xr.fresh` offer the possibility of rich feature sets for use in machine learning but it remains untested in the literature [42]. Despite their effectiveness, machine learning methods can be inconsistent [11]. Several factors contribute to this variability, including hyperparameter selection [12], sample quality and quantity [13], selection of training and testing splits [15], imbalanced classes [15], and feature generation and selection [14].

To address these limitations, deep learning approaches have gained traction in crop type mapping, with both Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) emerging as particularly promising architectures. RNNs are well suited to the sequential nature of remote sensing time series, as their internal hidden states allow them to retain a form of memory across time steps, capturing the temporal dynamics of crop phenology that static or summary-based features often miss. This capacity to model temporal structure has proven especially valuable for crop classification in data-rich environments, where the full growing season trajectory of spectral signals can be exploited [43, 29]. The lack of high quality training data in the Global South and lack of reliable cloud-free imagery has pushed many of the most successful efforts into the realm of precision agriculture such as applications for hyperspectral and drone-derived imagery [39]. Nevertheless, recent work applying deep learning to agricultural mapping in developing world contexts has demonstrated encouraging results. Ayushi et al. [45] applied a U-Net fully convolutional encoder-decoder architecture to a crop type mapping task in Kenya, and Liu et al. [46] used a convolutional neural network (3D-CNN) both achieving F1 scores over 73%. However questions remain about the generalizability of deep learning models particularly in complex environments with limited training data [47].

Hybrid approaches offer a compelling opportunity by combining the representational power of deep learning with the interpretability and efficiency of traditional machine learning, and several studies have demonstrated their value in agricultural remote sensing contexts. A common strategy in precision agriculture involves using a pre-trained deep learning model as a feature extractor especially, with the resulting high-level representations passed to a traditional classifier for the final prediction. Espejo-Garcia et al. [48] fine-tuned architectures including MobileNet, DenseNet, and

Xception before replacing their fully connected layers with Support Vector Machines and Gradient Boosting classifiers for weed identification, while Koklu et al. [49] found that pairing MobileNetV2 feature extraction with an SVM consistently outperformed end-to-end deep learning for grapevine leaf classification. Yang et al. [50] more relevant to this study, combined a CNN with a Random Forest classifier for Sentinel-2 crop type mapping, leveraging the spatial-spectral learning capacity of the CNN alongside the robustness and resistance to overfitting characteristic of ensemble tree methods. Of particular relevance to data-scarce environments, Nowakowski et al. [51] demonstrated that integrating transfer learning with traditional machine learning algorithms achieved high classification accuracy in Malawi, suggesting that hybrid architectures may be especially well suited to the constraints of the Global South. Taken together, these hybrid approaches suggest a productive middle path between the interpretability and data efficiency of classical machine learning and the feature-learning capacity of deep neural networks, and they form part of the methodological landscape against which the approach taken in this study is positioned.

This study addresses a critical yet underexplored gap in the crop classification literature: the systematic comparison of machine learning, deep learning, and hybrid approaches under the constraints typical of the Global South, but with the advantage of an unusually rich, high-quality training dataset. While much of the methodological literature has been developed and validated in data-rich temperate agricultural settings, the Western Cape of South Africa presents a distinct set of challenges — smaller field sizes, rain-fed dryland farming systems, heterogeneous and spatially variable planting rotations, persistent cloud cover during the critical winter growing season, and phenological cycles that do not conform to the assumptions embedded in many widely used models. We leverage this rare combination of environmental complexity and labeled data richness to conduct a rigorous evaluation of methods that are seldom compared within a single study and region. Our experiments utilize a labeled field-boundary dataset provided by Radiant Earth’s MLHub [3], capturing ground-truth crop types for agricultural fields in South Africa during the 2017 growing season.

We evaluate multiple deep learning architectures — including CNN-BiLSTM hybrids, TabNet ensembles, light-weight temporal-attention (L-TAE) and temporal-convolution (TempCNN) networks, and 3D CNN models — that extract dynamic phenological features across time, and benchmark these against traditional ensemble methods including XGBoost, LightGBM, and Random Forest. Critically, we evaluate every model twice: under the conventional protocol — random or field-wise cross-validation *within* the training region — and on a spatially disjoint holdout tile that no training pixel touches. This dual evaluation exposes a large and architecture-dependent generalization gap that single-region validation hides entirely, and it reorders the models. Throughout, we restrict our inputs to multi-temporal Sentinel-2 optical imagery alone, without recourse to SAR data, in order to isolate the signal available from spectral time series and maximize applicability to contexts where only optical data is available. All code is publicly available to support reproducibility and future research [4].

The specific contributions of this study are:

- We show that the **sparse, axis-aligned feature-selection inductive bias** behind the two-decade dominance of Random Forest and gradient-boosted trees in remote sensing is precisely the property that governs *spatial* transfer, and that TabNet inherits it (attentive sparsemax feature masks) while dense CNN/RNN/attention temporal nets discard it and overfit region-specific signal.
- We **quantify the in-region→spatial-holdout ranking inversion** across nineteen models: the dense temporal networks that lead in-region (and CNN-BiLSTM in particular, the model conventional evaluation singles out as best) become near-worst on the holdout, while parsimonious models transfer with the smallest gaps.
- We identify **field-level aggregation as a second, substitutable robustness lever** (variance reduction) that recovers transfer for dense temporal models but is redundant for already-sparse ones.
- We characterize **data efficiency** via a training-set reduction experiment, and provide the **first evaluation of `xr_fresh`** automated time-series feature extraction in crop classification.

This leads to the central research question: across a range of machine learning, deep learning, and hybrid architectures, which approach most reliably classifies crop types from multi-temporal Sentinel-2 optical imagery in a complex dryland farming environment *when evaluated on a spatially disjoint region*, and what properties of a model determine whether its in-distribution accuracy survives the transfer to unseen terrain?

1.1 Related Work

Crop classification using satellite imagery has become a core task in precision agriculture because it supports large-scale crop monitoring, yield estimation, resource management, and food-security analysis. The increasing availability of high-resolution, multi-temporal Earth observation data, particularly from Sentinel-2, has enabled substantial progress in crop mapping, but several challenges remain. These include strong spectral similarity among crop types, temporal variability in planting and phenological cycles, class imbalance, and the difficulty of building methods that generalize across regions and farming systems.

Early work in crop classification relied primarily on classical machine learning methods such as Random Forest, Support Vector Machines, Logistic Regression, and gradient-boosted tree models. These approaches typically operate on hand-crafted inputs derived from spectral bands, vegetation indices, temporal summaries, or other engineered descriptors, and they have often proven effective because they are computationally efficient, interpretable, and robust on structured remote-sensing data [28, 2]. However, their performance depends heavily on feature engineering and domain knowledge, which can limit transferability across crop systems and agroecological regions. This challenge becomes especially pronounced in heterogeneous agricultural landscapes, where differences in management practices, field conditions, and phenological timing can reduce the separability captured by manually designed features [28, 2].

These limitations have motivated growing interest in deep learning approaches that learn hierarchical feature representations directly from raw or minimally processed observations. Convolutional neural networks (CNNs) are effective at extracting local spectral and spatial patterns, while recurrent architectures such as LSTMs and BiLSTMs are well suited to modeling temporal crop-development signals over a season. More broadly, deep learning has become increasingly important in remote sensing because it reduces reliance on manually constructed predictors and can capture complex temporal structure that summary statistics often miss [29, 43]. Prior work has shown that temporal modeling is especially valuable when crop types are difficult to distinguish from single-date imagery but follow different seasonal trajectories [19, 22].

A particularly important direction in the literature has been the development of hybrid architectures that combine convolutional feature extraction with temporal sequence modeling. For example, Fan et al. proposed a hierarchical convolutional recurrent neural network for Sentinel-2 image classification, showing that combining CNN-based feature extraction with recurrent modeling can improve classification robustness relative to both classical and standalone deep learning baselines [6]. Similarly, Tufail et al. demonstrated that hybrid CNN-RNN models applied to multi-temporal Sentinel-2 imagery and red-edge vegetation indices can achieve very strong crop-mapping performance by jointly learning spatial context and temporal phenology [7]. Bandar and Coşkunçay likewise showed that combining an attention-based BiLSTM with a temporal CNN improves crop classification from Sentinel-2 time-series data, reinforcing the broader value of spatiotemporal representation learning in agricultural remote sensing [8].

Beyond fully end-to-end deep architectures, several studies have explored hybrid pipelines that combine learned representations with classical machine learning classifiers. Espejo-Garcia et al. used deep convolutional backbones as feature extractors before applying conventional classifiers for weed identification, while Koklu et al. found that deep features combined with an SVM outperformed end-to-end models for grapevine leaf classification [48, 49]. In a setting more directly related to this study, Yang et al. combined CNN-based representation learning with a Random Forest classifier for multi-temporal crop classification from Sentinel-2 imagery [50]. Transfer-learning-based hybrid approaches have also shown promise in data-constrained environments; for example, Nowakowski et al. reported strong crop-type mapping results in Malawi using transfer learning combined with traditional machine learning [51]. Taken together, these studies suggest that hybrid approaches can provide a useful middle ground between the flexibility of deep learning and the robustness and interpretability of classical models.

The representation of input data is another central theme in the literature. Sentinel-2 has become a preferred platform for crop monitoring because of its high revisit rate and multispectral richness, and these data are exploited either through engineered descriptors (spectral indices, temporal summaries) fed to classical learners or through end-to-end models that learn directly from temporal reflectance sequences. A recurring assumption is that richer, learned representations should generalize better than compact engineered ones; whether this holds under genuine spatial transfer — as opposed

to within-region validation — is less well established and is a central question of this study.

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region or scene [15]. Such protocols hold out individual pixels or fields but not *space*: training and test samples share the same atmospheric conditions, soils, management calendars, and sensor-acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational crop maps are produced precisely by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare [47]. This is compounded in the Global South, where rain-fed dryland farming, heterogeneous field conditions, irregular planting calendars, and persistent cloud cover make spatial transfer especially demanding [52, 39].

Our study is positioned within this gap. Rather than evaluating a single architecture in isolation, we conduct a systematic comparison of classical machine learning, deep learning, and patch-based approaches under *both* in-region field-wise validation and a spatially disjoint holdout, in the Western Cape of South Africa. We build on prior work in temporal modeling, hybrid learning, and ensemble-based classification, while also evaluating `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting [42]. This allows us to ask not only which approach is most accurate, but which model properties cause in-distribution accuracy to survive — or fail to survive — the move to unseen terrain.

2 Data Description

The dataset used for this study consists of two primary components: satellite imagery and field boundary data, as sourced from [20].

2.1 Study Area and Description

The study was conducted in an agriculturally productive region of the Western Cape of South Africa, characterized by diverse agroecological conditions and cropping systems. Major crop types in the dataset include wheat, barley, canola, small grain grazing, and lucerne/medics (alfalfa). The region spans varying climatic zones, from semi-arid to subtropical, and exhibits seasonal differences in rainfall and temperature, contributing to distinct crop phenologies.

The labeled data are organized into Sentinel-2 UTM grid tiles (zone 34S). Two adjacent tiles (34S_19E_258N and 34S_19E_259N) form the **training region**, comprising 4,150 labeled fields covering 622 km² of cultivated area (mean field size 15.0 ha, median 11.0 ha). A third, spatially disjoint tile (34S_20E_259N) is reserved as a **holdout region** for out-of-sample evaluation and contains 2,417 labeled fields over 282 km² (mean 11.7 ha, median 6.9 ha). The holdout tile shares no boundary pixels with the training tiles, so transfer to it tests generalization across soils, management practices, and acquisition geometry rather than merely across held-out fields

within a scene. Unless otherwise noted, “in-region” results refer to a field-wise (**fid**-stratified) train/validation/test split within the training region, while “out-of-sample” or “holdout” results refer to predictions on the disjoint tile.

Field parcels differ in size, shape, and management practices, introducing heterogeneity into the dataset. Sentinel-2 imagery was collected monthly across the 2017 growing season, covering key phenological stages from emergence to harvest. Each field carries a single crop-type label for the season. This temporal resolution is vital for capturing crop-specific growth patterns, which are often indistinguishable in single-date imagery.

2.2 Remote Sensing Data

Sentinel-2 Level-2A optical surface-reflectance data were acquired as monthly cloud-free composites over the study area in South Africa for the January–December 2017 growing season. Monthly median composites were generated using a 30% **s2cloudless** cloud-probability filter combined with a morphological cloud-shadow mask clipped to field boundaries, scaled to integer reflectance ($\times 10,000$), and exported as 10 m single-band GeoTIFFs via Google Earth Engine. From each monthly composite, six features were extracted: four optical bands — B2 (Blue, 490 nm), B6 (Red Edge 1, 740 nm), B11 (SWIR 1, 1610 nm), and B12 (SWIR 2, 2190 nm) — alongside the Enhanced Vegetation Index (EVI) and Hue, a colorimetric feature derived from spectral reflectance to characterize vegetation condition [9]. This band subset was chosen to span the spectral regions most informative for separating the target crops while limiting redundancy: the blue band (B2) constrains atmospheric and soil-brightness effects and enters the EVI formulation; the red-edge band (B6) is sensitive to chlorophyll content and canopy structure, which differ across cereals and broadleaf crops; and the two SWIR bands (B11, B12) respond to leaf and soil moisture and to senescence dynamics that help discriminate phenologically similar cereals such as wheat and barley (Figure 4). Red-edge and SWIR bands in particular are repeatedly identified as the most discriminating Sentinel-2 features for crop type in prior work [12, 14].

Each feature was captured across 11 temporal observations during the growing season; however, due to persistent cloud cover, data from May and June 2017 were excluded, yielding 10 valid monthly observations per feature. Each pixel was consequently represented by a 60-dimensional time-series feature vector, enabling the capture of rich phenological dynamics across the season.

EVI was computed from the monthly median composites, i.e. from per-band medians rather than from a single simultaneous acquisition. Because the EVI formula assumes its input bands are observed together, deriving it from independently composited bands can distort its absolute physical interpretation; the resulting feature is best read as a relative greenness descriptor that tracks seasonal phenology rather than as a calibrated instantaneous index. We retain it on that basis, and our feature-importance analysis (Section 4) confirms that the model leans far more heavily on the raw reflectance bands and their temporal statistics than on EVI, so this approximation does not drive the results. We flag the per-date band-consistency issue explicitly as a limitation (Section 5); pixel-wise temporal gap-filling that preserves simultaneity would be the appropriate remedy in work where a calibrated index is required.

By relying exclusively on optical data rather than SAR imagery [5], this study demonstrates the capability of multi-temporal spectral analysis alone for robust crop classification. This approach is directly transferable to emerging high-resolution optical platforms such as PlanetScope, reinforcing the operational relevance of dense, temporally-aware optical remote sensing for agricultural monitoring at scale.

2.3 Field Boundary Data

The field boundary polygons were drawn from the crop type classification dataset for Western Cape, South Africa [3]. This GeoJSON file delineated each agricultural field as a polygon, with boundaries digitized primarily from 50 cm aerial photography at scales of 1:2,500 for horticultural parcels and 1:7,500 for other fields between May and August 2017. In cases where ground reference was unavailable, Sentinel-2 imagery supplemented the digitizing process. Each polygon fully encompassed the field area and carried a crop-type label obtained via aerial and vehicle surveys conducted from May 2017 to March 2018 [3].

All predictions in our study were made at the field level.

2.4 Crop Types and Distribution

In this study, we classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing.

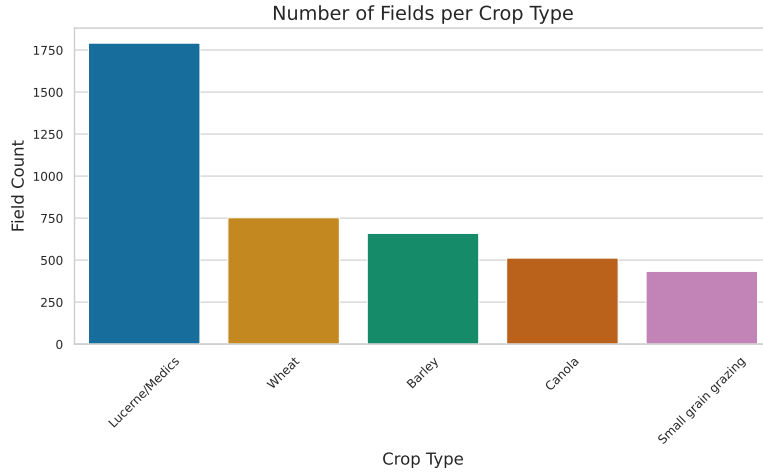


Fig. 1: Number of fields per crop type.

Figure 1 illustrates the distribution of crops in the study area. The data are notably imbalanced: in the training region lucerne/medics accounts for 1,792 of the 4,150 fields (43%), followed by wheat (753), barley (660), canola (512), and small grain grazing (433); the holdout region is more skewed still, with lucerne/medics comprising 1,360

of 2,417 fields (56%). This imbalance has important implications for model training and evaluation, as classification algorithms may be biased towards the majority class unless appropriate balancing techniques are applied, and it motivates our use of macro-averaged F1 as the primary metric (Section 3.3).

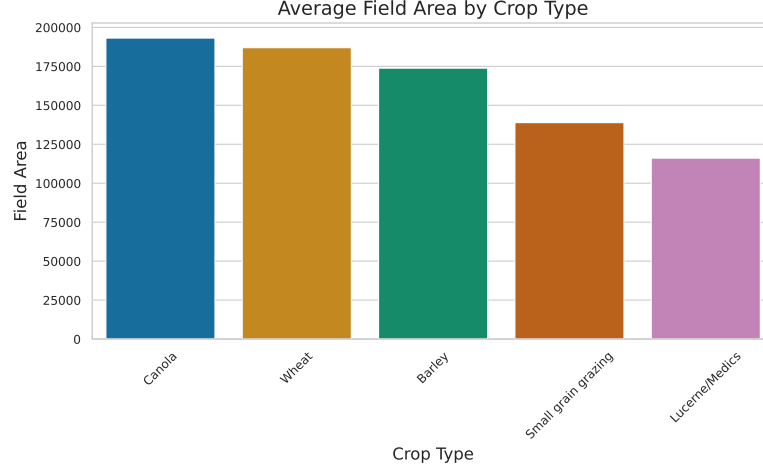


Fig. 2: Average field area by crop type (square metres, m²).

Figure 2 presents the mean field area for each crop type, measured in square metres (m²). Although lucerne/medics represent the majority of fields, their average area is smaller compared to other crop types.

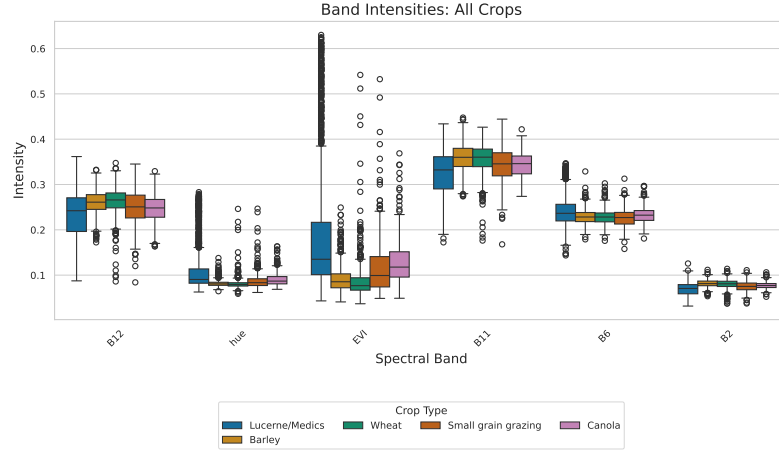


Fig. 3: Boxplots of spectral band intensities for each crop type, showing the distribution and overlap of spectral responses.

Figure 3 displays box plots of spectral band intensities for each crop type, illustrating the distribution and overlap of spectral responses. Although some differences in median and spread are evident, the substantial overlap in spectral intensities across crops demonstrates that classification based solely on a limited set of spectral bands is challenging and prone to confusion. This was particularly true for cereal crops such as wheat and barley, which exhibited highly similar spectral profiles, as shown in Figure 4. The close alignment of their spectral signatures underscores why models often struggle to distinguish between these two crops using spectral data alone [21].

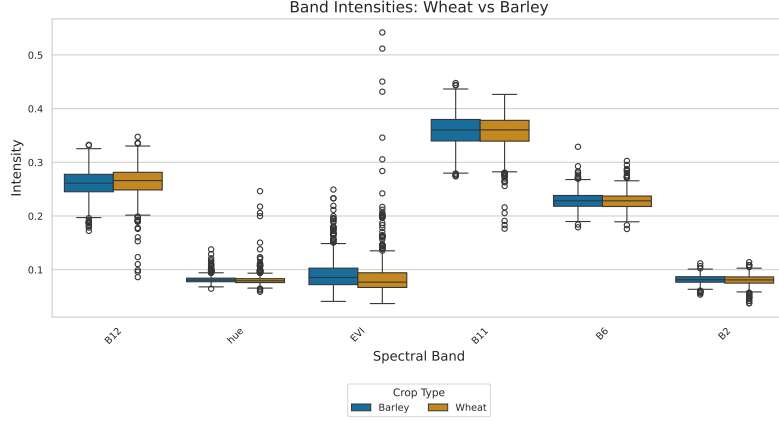


Fig. 4: Boxplots comparing spectral band intensities of wheat and barley, highlighting their high degree of spectral similarity and potential for classification confusion.

These findings highlight the need to incorporate temporal information that better represents phenological stages throughout the growing season to improve crop classification. The unique temporal dynamics associated with the stages of crop development (e.g., emergence, peak greenness, senescence) provide additional discriminatory power that can help resolve ambiguities between crops with similar spectral signatures [22]. Therefore, integrating both spectral and temporal data was crucial to enhance classification accuracy, especially in complex agricultural landscapes where crops like wheat and barley were spectrally similar.

3 Methodology

This study explores three complementary modeling paradigms for crop classification: feature-based machine learning, deep learning, and patch-based approaches. These paradigms differ in how spatial and temporal information from multi-spectral Sentinel-2 imagery is represented and utilized.

Classical machine learning relies on engineered features derived from pixel-level time series, while deep learning models learn hierarchical representations directly from raw data. In contrast, patch-based approaches operate on spatially contiguous regions,

enabling the model to capture local spatial structure and contextual relationships that are not available in purely pixel-wise representations. Each paradigm was evaluated at both the pixel level and the field (or patch) level to assess their effectiveness across different representations of the data.

Crop classification from remote sensing data can therefore be approached through three complementary strategies: feature engineering, representation learning, and spatial context modeling.

In the feature engineering paradigm, domain knowledge is used to construct descriptive inputs from raw spectral time series. These include vegetation indices such as NDVI, spectral band ratios, textural measures, and phenological statistics, which are then used as inputs to classical machine learning models such as Random Forests (RF), Support Vector Machines (SVM), and Gradient Boosted Trees (GBT) [2]. In this study, we adopt the `xr_fresh` framework to automatically extract a comprehensive set of time-series features, enabling the construction of high-dimensional representations without manual feature design [42].

In contrast, deep learning follows a representation learning paradigm, where models learn hierarchical feature representations directly from raw input data. Convolutional Neural Networks (CNNs) capture local spatial and spectral patterns, while Recurrent Neural Networks (RNNs) and Transformer-based architectures model temporal dependencies and long-range interactions in multi-temporal data [17]. These approaches reduce reliance on hand-crafted features and allow the model to learn complex crop growth dynamics directly from the data.

Beyond pixel-level modeling, patch-based approaches introduce an additional spatial dimension by processing contiguous regions of imagery rather than individual pixels. This enables models to capture local texture, spatial heterogeneity, and contextual relationships between neighboring pixels, which are particularly important in agricultural landscapes where crop patterns exhibit strong spatial coherence. Patch-based models therefore complement both classical and deep learning approaches by explicitly incorporating spatial context into the classification process.

This study systematically compares these paradigms to evaluate their relative strengths, computational trade-offs, and effectiveness in handling complex agricultural environments, particularly under conditions of spectral similarity and temporal variability.

Model performance was assessed using a combination of standard classification metrics, including Cohen’s Kappa, F1-score, and prediction entropy. These metrics were selected to evaluate not only overall accuracy but also class balance and prediction confidence across imbalanced crop types.

3.1 Classical Machine Learning

In the classical approach, we automated feature extraction using the `xr_fresh` toolkit to rapidly compute a comprehensive set of statistical and temporal features from each 11-month spectral time series [9, 42]. To address class imbalance, we applied SMOTE. Several classifiers such as Logistic Regression, Random Forest, LightGBM, and XGBoost were then evaluated.

We extracted the following features from each pixel’s 11-month spectral profile [9]:

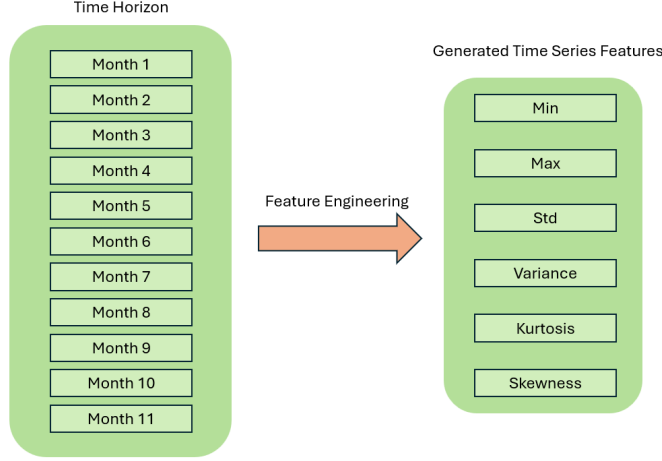


Fig. 5: Extraction of statistical and temporal features from the 11-month spectral time series.

- **Absolute Energy:** Sum of squared values, capturing overall signal magnitude.
- **Absolute Sum of Changes:** Total absolute difference between consecutive measurements, reflecting temporal variability.
- **Autocorrelation (1,2-month lag):** Correlation with its one- and two-month lagged versions, indicating series persistence.
- **Count Above Mean:** Number of points above the series mean, quantifying high-value occurrences.
- **Day of Year of Max/Min:** Julian day when the maximum and minimum occur, marking peak and low activity periods.
- **Kurtosis:** Tail heaviness of the distribution, highlighting propensity for extremes.
- **Linear Trend:** Slope of a least-squares fit, summarizing the overall upward or downward trend.
- **Longest Strike Above/Below Mean:** Longest consecutive run above or below the mean, capturing sustained periods.
- **Max/Min:** Absolute extreme values in the series.
- **Mean/Median:** Measures of central tendency.
- **Mean Absolute Change/Mean Change:** Average magnitude and signed change between consecutive points.
- **Mean Second Derivative:** Mean of the series' second differences, detecting acceleration or deceleration.
- **Quantiles (0.05, 0.95):** Values at the 5th and 95th percentiles, summarizing distribution extremes.
- **Ratio Beyond $r\sigma$:** Proportion of points more than r standard deviations ($r=1,2,3$) from the mean.
- **Skewness:** Asymmetry of the distribution around the mean.
- **Standard Deviation/Variance:** Spread measures around the mean.

- **Sum of Values:** Total sum of observations (mean $\times$ length).
- **Symmetry Looking:** Similarity when the series is reversed.
- **Time Series Complexity:** Complexity-Invariant Distance metric of sequence irregularity.

This toolkit allowed us to extract all required time-series features efficiently, ensuring each pixel’s temporal signature was well-represented for robust classification [42].

3.1.1 Pixel-Level Analysis

For pixel-level analysis, each pixel was treated as an independent sample. Feature vectors from each pixel’s 11-month history were used to train classifiers. During inference, pixel predictions within a field polygon were aggregated via majority voting to determine the overall field label.

Baseline Classical Models:

To benchmark crop classification, we used four classical models: Logistic Regression, Random Forest, LightGBM, and XGBoost—ranging from interpretable linear models to advanced tree-based learners.

Each pixel was represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^{174}$, formed from time-series features across ten months and six spectral indices (B2, B6, B11, B12, EVI, Hue), capturing phenological signatures over time.

These four models are standard and we summarize only their configurations here; their formal definitions are collected in Appendix A. **Logistic Regression** [23] is a linear multinomial classifier providing interpretable per-class weights. **Random Forest** [24] is a bagged ensemble of decision trees whose axis-aligned splits capture non-linear interactions; we used 300 trees. **LightGBM** [25] and **XGBoost** [26, 27] are gradient-boosted tree ensembles with leaf-wise growth and explicit L_1/L_2 regularization; both were trained with GPU histogram methods. All four operate on the same per-field `xr_fresh` feature vector and, being tree- or margin-based with built-in feature selection or shrinkage, serve as the parsimonious baselines against which the deep models are compared.

Pixel-Level Ensemble with Hybrid Voting:

We ensembled RF, XGBoost, and LightGBM using soft voting. Per-pixel class probabilities were averaged:

$$\hat{\mathbf{P}}_i = \frac{1}{3} (\mathbf{P}_{\text{RF}}(\mathbf{x}_i) + \mathbf{P}_{\text{XGB}}(\mathbf{x}_i) + \mathbf{P}_{\text{LGBM}}(\mathbf{x}_i)) \quad \hat{y}_i = \arg \max_k \hat{P}_i(k)$$

Field-Level Aggregation:

Let field f consist of pixels $i \in f$. Field-level soft voting is computed as:

$$\bar{\mathbf{P}}_f = \frac{1}{|f|} \sum_{i \in f} \hat{\mathbf{P}}_i \quad \hat{y}_f^{\text{soft}} = \arg \max_k \bar{P}_f(k)$$

We introduced a confidence margin:

$$\Delta_f = \bar{P}_f^{(1)} - \bar{P}_f^{(2)}$$

where $\bar{P}_f^{(1)}$ and $\bar{P}_f^{(2)}$ are the highest and second-highest class probabilities in $\bar{P}_f$, respectively.

If $\Delta_f < \theta$ (threshold = 0.10), we fall back to mode voting across pixel labels:

$$\hat{y}_f^{\text{mode}} = \text{mode} \{ \hat{y}_i \mid i \in f \}$$

Final Hybrid Rule:

$$\hat{y}_f = \begin{cases} \hat{y}_f^{\text{soft}}, & \text{if } \Delta_f \geq \theta \\ \hat{y}_f^{\text{mode}}, & \text{otherwise} \end{cases}$$

This strategy balanced probabilistic confidence with categorical consistency, improving field-level classification stability.

3.1.2 Field-Level Analysis

In the field-level analysis, we aggregated all pixel features within each field polygon, as shown in Figure 6, by computing the mean across pixels to form a single representative feature vector per field. This step reduced noise and the influence of outlier pixels while preserving meaningful spatial patterns. Classifiers were then trained using these aggregated field-level vectors to directly learn from spatially coherent crop representations. This approach aligns with previous findings that object-based aggregation helps in minimizing spectral confusion and enhances classification accuracy, especially in homogeneous agricultural regions [28].

To address class imbalance at the field level, we combined **SMOTETomek** resampling with a **stacked ensemble classifier**. This approach ensured both balanced data and strong generalization performance.

SMOTETomek Resampling

SMOTETomek combines oversampling and undersampling techniques. SMOTE generates synthetic samples for underrepresented classes by interpolating between existing samples, while Tomek Links removes borderline examples to improve class separability. This helped mitigate the severe class imbalance visible in Figure 1, especially for crops like small grain grazing and canola.

Stacked Ensemble Classifier

We built a two-layer ensemble consisting of Random Forest and XGBoost as base learners, with Logistic Regression (using `passthrough=True`) as the meta-learner,

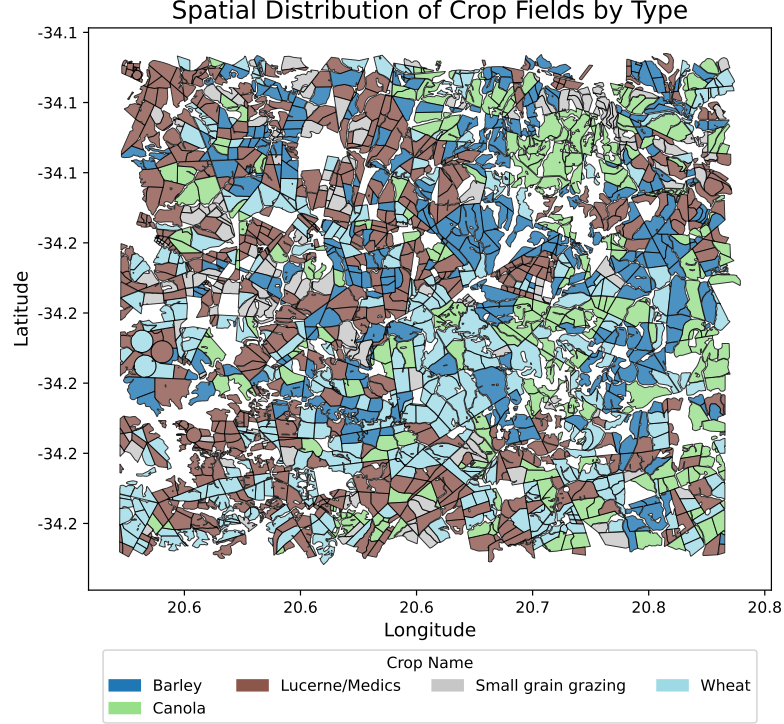


Fig. 6: Illustration of field-level aggregation: pixel values are grouped by field polygon and summarized into a single feature vector.

allowing it to use both base-model predictions and the original input features. The final prediction for pixel i is defined as:

$$\hat{y}_i^{\text{stacked}} = f_{\text{meta}}(f_{\text{RF}}(\mathbf{x}_i), f_{\text{XGB}}(\mathbf{x}_i), \mathbf{x}_i)$$

This architecture combines the non-linear modeling capacity of tree-based learners with a linear meta-learner that integrates both base predictions and original input features.

Field-Based Aggregation and XGBoost Optimization

We also implemented an optimized field-level pipeline using XGBoost. Field identifiers were used to ensure that no field contributed to more than one data split, with fields split into train/validation/test sets using `fid`-based stratification. Features were aggregated via the mean across pixels, and labels via the mode crop type per field.

Optuna was then used to tune XGBoost over 75 trials, searching across `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `gamma`, `reg_alpha`, `reg_lambda`, and `min_child_weight`, with the objective of maximizing the

weighted F1 score via 5-fold stratified cross-validation. The best configuration (Trial 32) is detailed in Table 1, which also achieved a validation accuracy of 76.61%.

Table 1: Top 5 trials from hyperparameter search with corresponding parameters and validation accuracies

Trial	Acc.	Est.	Depth	LR	SubS	ByTree	Gamma	Reg α	Reg λ	MinCW
32	76.61%	1141	13	0.0215	0.5749	0.6986	0.5589	0.00184	0.00259	5
12	76.46%	984	12	0.0145	0.6675	0.6946	0.0799	0.00503	0.00455	7
24	76.40%	1165	11	0.0162	0.7831	0.6403	0.00413	0.00481	0.00478	9
13	76.29%	995	12	0.0277	0.6345	0.6829	0.3999	0.00011	0.00567	6
18	76.29%	915	13	0.0121	0.6784	0.5958	0.1612	0.00491	0.1732	8

3.2 Deep Learning Architectures for Crop Classification

Deep learning models ingest raw spectral values and learn hierarchical representations that capture both spatial and temporal dependencies, offering an alternative to the manual feature engineering required by classical approaches [28, 29, 30]. We explore architectures across two spatial granularities: pixel-level models that operate on per-pixel temporal sequences, and patch-level models that additionally exploit local spatial structure. To ensure unbiased evaluation across all architectures, all pixels belonging to a given field were assigned exclusively to either the training, validation, or test set, preventing spatial information leakage between splits.

3.2.1 Pixel-Level Architectures

At the pixel level, each sample is an individual pixel characterized by 60 temporal features spanning the 2017 growing season — six spectral bands and vegetation indices (B2, B6, B11, B12, EVI, Hue) observed across 10 monthly composites. Each pixel is linked to a unique field identifier and labeled according to the field crop type across five classes: wheat, barley, canola, small grain grazing, and lucerne/medics. Pixel-level predictions are subsequently aggregated to the field level via majority voting:

$$\hat{y}_f = \text{mode}(\{\hat{y}_i \mid i \in f\})$$

For TabNet models we applied z-score standardization; CNN-based models were trained on raw reflectance values, allowing convolutional layers to learn directly from temporal magnitude patterns.

CNN + BiLSTM Ensemble with Focal Loss.

To capture the intricate temporal dynamics of crop growth, we developed a hybrid architecture combining 1D Convolutional Neural Networks (CNNs) with Bidirectional Long Short-Term Memory (BiLSTM) networks (Figure 7).

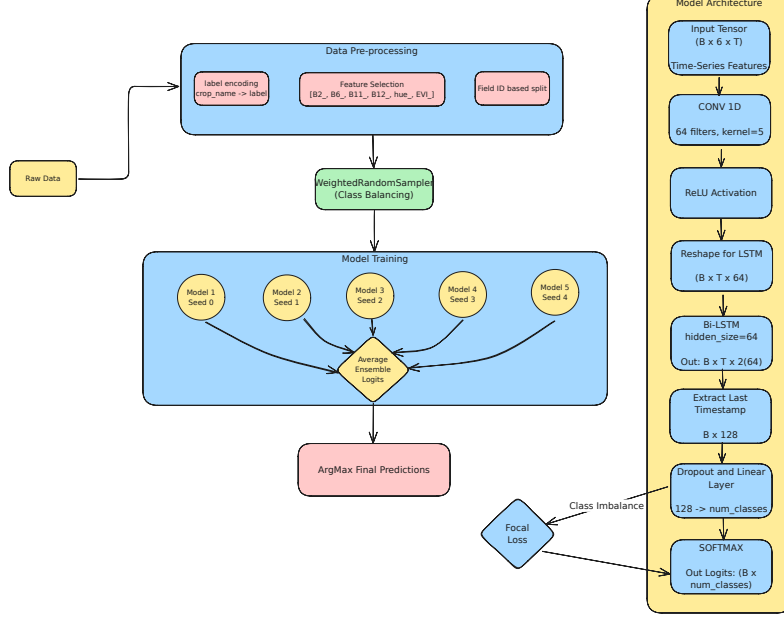


Fig. 7: Architecture of the CNN + BiLSTM model. The model combines 1D convolutional temporal feature extraction with sequential modeling through a BiLSTM layer.

The model processes an input tensor of shape $(B, 6, T)$. A 1D convolutional layer with 64 filters and kernel size 5 is applied along the temporal axis, followed by ReLU activation and tensor permutation to $(B, T, 64)$ for compatibility with the recurrent layer. A bidirectional LSTM with hidden size 64 then captures temporal dependencies in both forward and backward directions, yielding an output of shape $(B, T, 128)$. The representation at the final timestep passes through dropout and a fully connected layer, with class probabilities computed via softmax:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{e^{z_{i,c}}}{\sum_{c'=1}^C e^{z_{i,c'}}}$$

To address class imbalance we used Focal Loss:

$$\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where α_t is the class weighting factor and $\gamma = 2.0$ is the focusing parameter. A **WeightedRandomSampler** further ensured balanced class representation during training. To reduce variance, five identical models were trained with different random seeds and their logits averaged:

$$\text{Logits}_{\text{ensemble}} = \frac{1}{N} \sum_{k=1}^N \text{Logits}^{(k)}$$

Ensemble of TabNet Models.

TabNet treats satellite observations as structured tabular inputs — each pixel’s repeated measurements organized as a flat feature vector — and learns through iterative attention-based feature selection rather than temporal ordering (Figure 8). At each decision step, an Attentive Transformer generates a sparse mask selecting the most relevant input dimensions, which are then transformed by a Feature Transformer. This enables the model to focus on different feature subsets at different depths without relying on explicit temporal structure [33, 32].

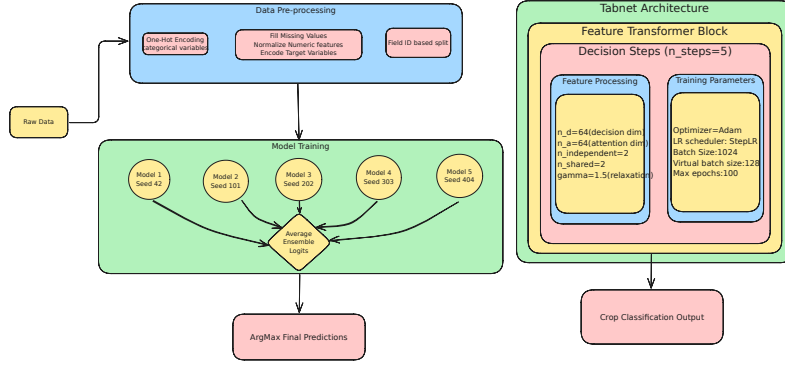


Fig. 8: Architecture of the TabNet ensemble. Each model applies iterative attention-based feature selection. Final predictions are obtained by averaging softmax outputs across five models, followed by field-level majority voting.

The network used $n_d = n_a = 64$, independent and shared block widths of 2, relaxation parameter $\gamma = 1.5$, and 5 decision steps. Training used the Adam optimizer with a StepLR scheduler, batch size 1024, virtual batch size 128, and early stopping monitored on a field ID-stratified validation set.

An ensemble of five TabNet models, each initialized with a distinct random seed (42, 101, 202, 303, 404), averaged softmax outputs to yield final class probabilities:

$$\hat{\mathbf{p}}_{\text{ensemble}} = \frac{1}{5} \sum_{i=1}^5 \text{Softmax}(\mathbf{z}^{(i)}), \quad \hat{y} = \arg \max_c \hat{p}_{\text{ensemble},c}$$

A practical advantage of TabNet is interpretability: the sparse decision masks can be inspected post hoc to identify which spectral bands or vegetation indices most influenced each prediction [32].

1D CNN with Optuna Hyperparameter Tuning.

As a lightweight baseline we implemented a 1D CNN whose architecture was optimized via Optuna [36] (Figure 9). The input is reshaped to $(B, 6, T)$, representing six spectral channels across monthly intervals, and passed through a single convolutional

layer followed by ReLU, adaptive average pooling, dropout, and a fully connected classification head. The hyperparameter optimization objective was:

$$\theta^* = \arg \max_{\theta \in \mathcal{H}} \mathcal{A}_{\text{val}}(f_{\theta})$$

where $\theta = \{\text{lr, dropout, filters, kernel size}\}$, solved using the Tree-structured Parzen Estimator (TPE). Each trial was evaluated over 25 epochs with a class-weighted sampler. The best configuration (Table 2) used a kernel size of 7, 64 filters, dropout ≈ 0.49 , and learning rate 0.00116, yielding a validation accuracy of 75.5%.

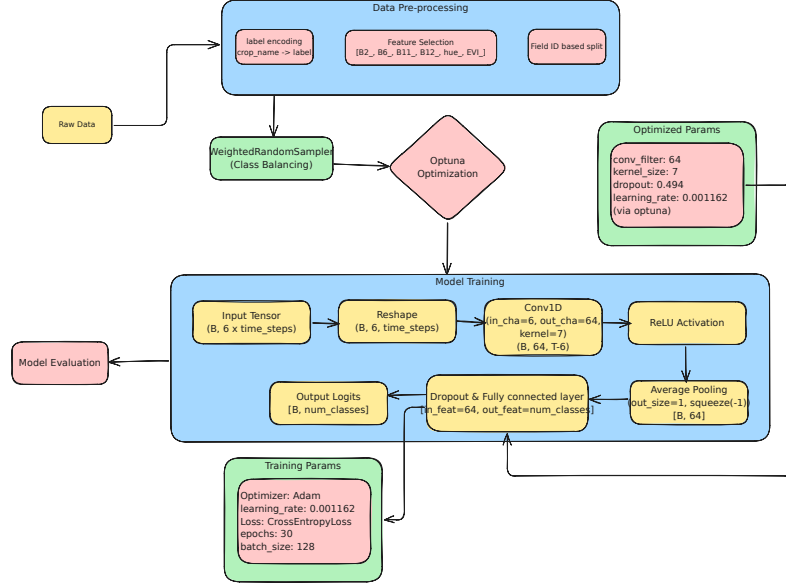


Fig. 9: Architecture of the 1D CNN model optimized via Optuna: a single convolutional layer followed by ReLU activation, adaptive pooling, and a fully connected classification head.

Table 2: Top Optuna trials for the 1D CNN ranked by validation accuracy.

Trial	Val. Acc.	LR	Dropout	Filters	Kernel
14	75.53%	0.00116	0.5	64	7
13	74.06%	0.00102	0.5	128	7
19	74.61%	0.00118	0.4	64	7
7	73.20%	0.00013	0.3	128	7
11	73.66%	0.00006	0.4	128	7

Lightweight Temporal Attention Encoder (L-TAE).

The L-TAE applies self-attention directly to the temporal sequence of per-pixel reflectance vectors, learning which observations in the season are most informative [34]. Given the embedded sequence $\{\mathbf{z}_t\}_{t=1}^T$, a set of learned master query vectors attend over time; for a single head the output is a temporally weighted sum

$$\mathbf{o} = \sum_{t=1}^T \alpha_t \mathbf{v}_t, \quad \alpha_t = \frac{\exp(\mathbf{q}^\top \mathbf{k}_t / \sqrt{d})}{\sum_{t'=1}^T \exp(\mathbf{q}^\top \mathbf{k}_{t'} / \sqrt{d})},$$

where $\mathbf{k}_t, \mathbf{v}_t$ are key/value projections of $\mathbf{z}_t$, $\mathbf{q}$ is the learned query, and d is the head dimension. Channel grouping keeps the encoder lightweight. We evaluated L-TAE both at the pixel level and at the field level (averaging pixel sequences within a field before encoding), and trained a five-seed ensemble.

Temporal Convolutional Network (TempCNN).

TempCNN classifies a pixel from stacked one-dimensional convolutions applied along the temporal axis, each capturing local phenological motifs of increasing receptive field [35]. A convolutional layer computes

$$h_t^{(l)} = \text{ReLU}\left(\text{BN}\left(\sum_{\tau=-k}^k W_\tau^{(l)} h_{t+\tau}^{(l-1)} + b^{(l)}\right)\right),$$

with batch normalization (BN) and dropout for regularization, followed by a fully connected softmax head. As with L-TAE, we evaluated pixel- and field-level variants and ensembled across five seeds.

3.2.2 Patch-Level Architectures

To exploit local spatial structure, we transitioned to a patch-level analysis in which each field is divided into a grid of fixed-size square patches, each retaining the full 10-month spectral history for every pixel (Figure 10). Compared with pixel-level models, this approach encodes spatial relationships between neighbouring pixels and provides implicit data augmentation by generating multiple training samples per field. The trade-off is increased computational cost and the omission of boundary pixels that fall outside the patch grid.

Example of Generated Patches by Field

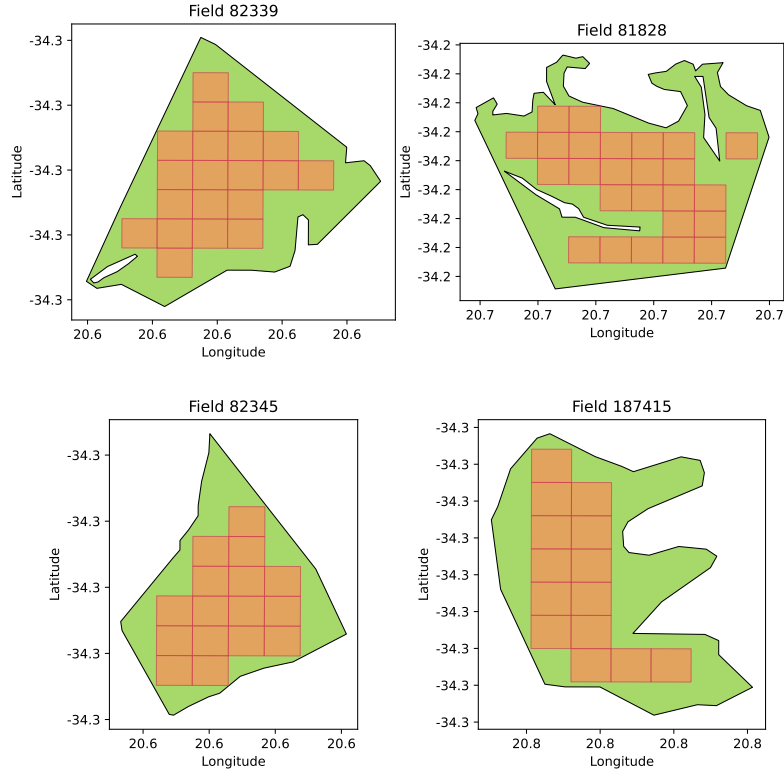


Fig. 10: Illustration of patch generation per field.

We investigated three distinct patch-based architectures.

Multi-Channel 2D CNN.

Each spectral band and monthly observation is treated as a separate input channel, giving a patch tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ ($H = W = 128$, $C = 66$). Two Conv2D blocks with K filters each compute:

$$Y_k[i, j] = \text{ReLU} \left(\sum_{c=1}^C \sum_{m=-1}^1 \sum_{n=-1}^1 W_{k,c,m,n} \cdot X[i + m, j + n, c] + b_k \right)$$

The feature map is flattened to $\mathbf{z} \in \mathbb{R}^{128}$ and passed to a fully connected classification head:

$$\hat{\mathbf{p}} = \text{softmax}(W_{\text{fc}} \mathbf{z} + b_{\text{fc}})$$

Ensemble of Class-Specific 3D CNNs.

Five class-specific 3D CNNs are trained, each producing a scalar score $p_i = f_i(\mathbf{X})$ for its target class. A logistic regression meta-learner combines these scores into a final prediction (Figure 11):

$$\hat{\mathbf{p}} = \text{softmax}(W_{\text{meta}}[p_1, \dots, p_5]^\top + b_{\text{meta}})$$

By modelling volumetric relationships jointly across spectral bands, spatial dimensions, and time, 3D convolutions capture crop-specific texture and phenological trajectories that 2D methods treat independently.

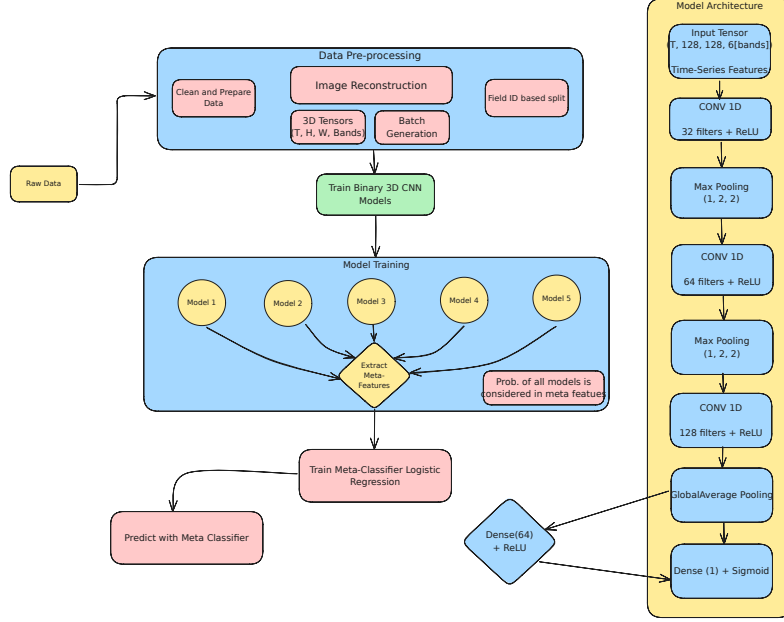


Fig. 11: 3D CNN ensemble architecture for patch-level crop classification.

Transformer-Based Model.

Each patch is represented as a sequence of multi-temporal image tensors of shape (T, H, W, B) , where T is the number of temporal observations, $H \times W$ is the spatial resolution (128×128), and B is the number of spectral bands and indices.

For each time step, a lightweight convolutional encoder is applied using a 1×1 convolution followed by global average pooling to produce a compact feature vector $\mathbf{z}_t \in \mathbb{R}^d$, where $d = 32$ in our implementation. This results in a temporal sequence of patch-level embeddings $\{\mathbf{z}_t\}_{t=1}^T$.

To incorporate temporal order, learned positional embeddings are added:

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t + \mathbf{e}_t$$

The resulting sequence is processed using stacked Transformer encoder blocks, each consisting of multi-head self-attention, residual connections, dropout, and a position-wise feedforward network:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

where $Q = \mathbf{Z}W_Q$, $K = \mathbf{Z}W_K$, and $V = \mathbf{Z}W_V$.

Finally, the encoded sequence is aggregated via global average pooling across time and passed through a fully connected layer with softmax activation to produce class probabilities:

$$\hat{\mathbf{p}} = \text{softmax}(W \cdot \mathbf{h} + b)$$

This architecture emphasizes temporal dependency modeling across monthly patch representations, while spatial information is incorporated indirectly through the convolutional preprocessing stage.

3.3 Performance Metrics

Because the class distribution is severely imbalanced (Figure 1), plain accuracy is misleading — a classifier can score highly simply by predicting the majority class. We therefore adopt **macro-averaged F1** ($F1_m$), the unweighted mean of the per-class F1 scores, as our *primary* metric: it weights every crop equally and so directly penalizes failure on the minority cereals, which is exactly where models differ. We report three further metrics alongside it: **Cohen’s Kappa**, chance-corrected agreement between predicted and true labels; **weighted F1**, the support-weighted mean of per-class F1; and **log-loss (cross-entropy)**, the mean negative log-probability assigned to the true class (lower is better, capturing prediction confidence). Higher Kappa and F1 indicate better, more balanced classification. Formal definitions of all four metrics are given in Appendix B.

4 Results

We evaluated every model under two protocols. *In-region* performance uses a field-wise (**fid**-stratified) train/validation/test split drawn entirely from the training tiles — the conventional way crop classifiers are benchmarked. *Out-of-sample* (OOS) performance applies the trained model to the spatially disjoint holdout tile (34S.20E.259N; 2,417 fields). All predictions are aggregated to the field level. Because the classes are heavily imbalanced (Section 3.3), we report **macro-averaged F1** ($F1_m$) as the primary metric, with Cohen’s Kappa, weighted F1, and log-loss (entropy) alongside.

4.1 In-region validation overstates performance and misranks models

Table 3 reports both protocols for all individual models, ranked by out-of-sample macro-F1, with the generalization gap $\Delta = F1_m^{\text{OOS}} - F1_m^{\text{in-region}}$ shown alongside; Figure 12 visualizes the same data ordered by Δ . Two facts stand out. First, *every* model loses accuracy out of sample, but the loss is highly uneven: it ranges from -0.05 (logistic regression) to -0.26 (CNN-BiLSTM). Second, this unevenness **reorders the models**. Under in-region validation the dense temporal deep nets look strongest — CNN-BiLSTM, TempCNN, and L-TAE all reach $F1_m \approx 0.72$ – 0.78 , and conventional field-level evaluation (on which the original analysis of this dataset relied) singled out CNN-BiLSTM as the best model. On the holdout tile that conclusion collapses: CNN-BiLSTM falls to $F1_m = 0.46$ (Kappa 0.32), the *largest* drop of any model and second-worst absolute score. The same fate befalls the 3D CNN patch ensemble (-0.22) and SMOTE-resampled stacking (-0.21).

Table 3: In-region (field-wise CV) versus out-of-sample (spatial holdout) macro-F1 for every individual model, ranked by out-of-sample macro-F1 — the metric we argue model selection should use. In-region macro-F1 and the gap Δ are shown alongside (Figure 12 orders the same data by Δ); OOS Kappa, weighted F1 (wF1), and cross-entropy (Xent) complete the picture. The best transferer is a TabNet pixel ensemble, whereas the model conventional field-level evaluation singles out as best (CNN-BiLSTM) is among the worst; the highest in-region scores (L-TAE, TempCNN) do not yield the highest out-of-sample scores.

Model	Level	In-reg $F1_m$	OOS $F1_m$	Δ	OOS κ	OOS wF1	OOS Xent
TabNet	pixel	0.73	0.60	-0.13	0.54	0.70	4.42
L-TAE	field	0.67	0.59	-0.07	0.47	0.67	5.51
L-TAE	pixel	0.78	0.58	-0.20	0.49	0.68	5.20
XGBoost	field	0.70	0.56	-0.13	0.47	0.68	4.89
Voting	field	0.69	0.56	-0.12	0.47	0.66	4.82
TempCNN	pixel	0.77	0.56	-0.21	0.49	0.68	5.09
Base LightGBM	pixel	0.62	0.56	-0.06	0.47	0.66	4.85
Logistic Regression	pixel	0.61	0.56	-0.05	0.47	0.67	4.93
LightGBM	field	0.68	0.56	-0.12	0.45	0.66	4.99
Stacking	field	0.64	0.55	-0.09	0.47	0.67	5.01
Random Forest	pixel	0.65	0.55	-0.10	0.45	0.65	5.07
Base XGBoost	pixel	0.64	0.54	-0.09	0.47	0.66	5.01
TempCNN	field	0.67	0.54	-0.14	0.43	0.64	6.07
SMOTE Stacked	field	0.69	0.49	-0.21	0.43	0.63	5.37
3D CNN	patch	0.70	0.48	-0.22	0.30	0.55	7.79
CNN-BiLSTM	pixel	0.72	0.46	-0.26	0.32	0.56	7.73
TabNet Temporal	field	0.62	0.43	-0.19	0.31	0.56	7.35
TabNet (xr_fresh)	field	0.51	0.37	-0.15	0.25	0.53	6.13

The winners under spatial transfer are a different group entirely. The best single model is the **TabNet pixel ensemble** ($F1_m = 0.60$, Kappa 0.54), and the most

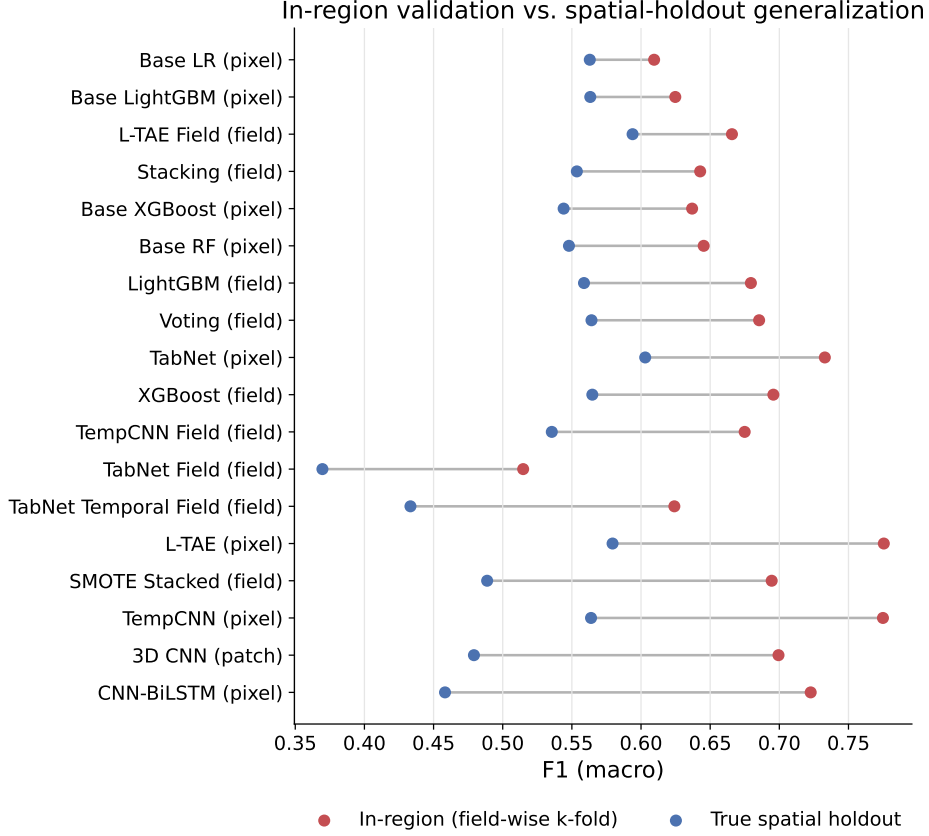


Fig. 12: In-region (field-wise cross-validation) versus spatial-holdout macro-F1 for each model, sorted by the gap. Dense temporal/patch deep nets (top) suffer the largest degradation; tree-based, linear, and field-aggregated models (bottom) transfer with the smallest gaps.

robust models — those with the smallest gaps — are classical: logistic regression ($\Delta = -0.05$), pixel-level LightGBM (-0.06), and the tree-based field ensembles. A majority-vote ensemble over all models reaches $F1_m = 0.60$ (Kappa 0.52), essentially tying TabNet but not exceeding it, indicating that voting cannot rescue the fragile members. The practical implication is immediate: a practitioner who selected a model by in-region cross-validation would have deployed the single worst transferer.

4.2 Pixel versus field level: aggregation as variance reduction

Several architectures were run at both the pixel level and the field level (averaging the per-pixel time series within each field before classification). Table 4 and Figure 13 show that field-level aggregation acts as a variance-reduction regularizer for the *dense* temporal networks: L-TAE improves from $F1_m = 0.58$ (pixel) to 0.59 (field) while

its gap shrinks from -0.20 to -0.07 , and TempCNN’s gap narrows from -0.21 to -0.14 . Averaging pixels within a field cancels region-specific pixel noise before the model sees it, lowering the in-region ceiling but improving transfer. **TabNet is the informative exception:** aggregation *hurts* it badly ($F1_m$ falls from 0.60 to 0.37), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel sample volume it exploits (millions of pixels versus a few thousand fields). The tree ensembles transfer well at either level. In short, dense models need aggregation to generalize; sparse, tree-like models do not.

Table 4: Out-of-sample macro-F1 (and gap Δ) at pixel versus field level for architectures evaluated at both granularities. Field aggregation helps dense temporal nets and the tree models, but degrades TabNet.

Architecture	Pixel		Field	
	OOS $F1_m$	Δ	OOS $F1_m$	Δ
L-TAE	0.58	-0.20	0.59	-0.07
TempCNN	0.56	-0.21	0.54	-0.14
TabNet	0.60	-0.13	0.37	-0.15
XGBoost	0.54	-0.09	0.56	-0.13
LightGBM	0.56	-0.06	0.56	-0.12

4.3 Data efficiency under spatial transfer

To probe how much labeled data the transferable models actually require, we retrained a representative set on random subsets of 75%, 50%, and 25% of the training fields and re-evaluated on the holdout tile (Figure 14). The most robust models are also the most data-efficient: TabNet retains essentially full holdout accuracy down to a quarter of the fields ($F1_m = 0.60, 0.62, 0.62, 0.59$ at 100/75/50/25%), and logistic regression is similarly flat (0.56–0.60). The dense pixel L-TAE degrades more steeply (0.58 to 0.57), and the explicit feature-selection network LassoNet is poor at every budget ($F1_m \approx 0.46$). Holdout accuracy is therefore limited far more by spatial transfer than by training-set size for the robust models — halving the labels costs them little.

4.4 Feature dimensionality and redundancy

A natural concern with the 174-dimensional `xr.fresh` representation is feature redundancy. We tested three forms of explicit dimensionality control. *LassoNet*, which learns a hard feature-selection mask, generalized worst of all pixel models ($F1_m \approx 0.46$ at every training budget; Figure 14), indicating that aggressive pruning discards transfer-relevant signal. A *sparse-attention* L-TAE variant (TabNet-style channel gating) modestly improved transfer over the dense L-TAE at matched data (0.61 vs. 0.57 at 75% of fields), consistent with *soft* sparsity helping. *L₂ regularization* of the tree

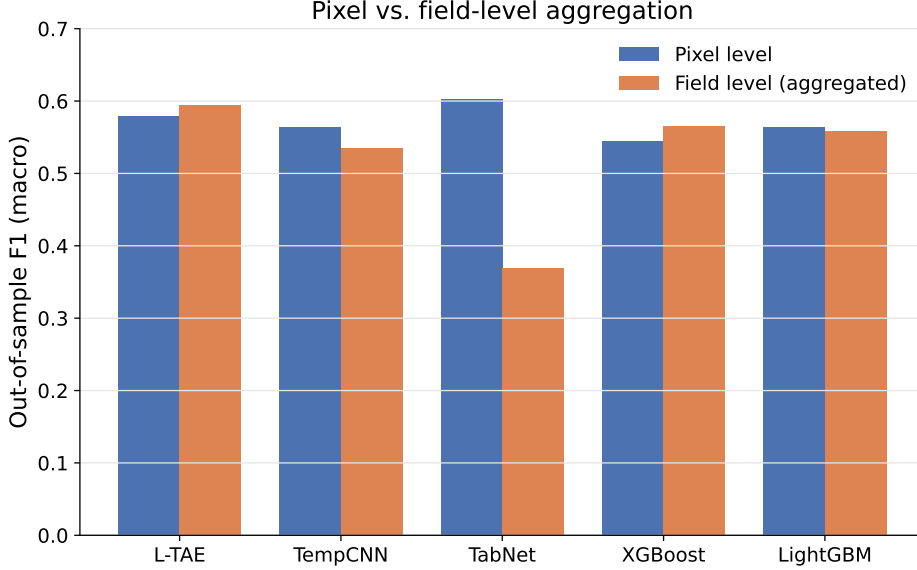


Fig. 13: Out-of-sample macro-F1 at pixel versus field level. Field-level aggregation recovers transfer for dense temporal models (L-TAE) but collapses the sparsity-reliant TabNet.

models changed OOS $F1_m$ negligibly (within ± 0.01). Together these show that the high feature dimensionality is not the binding constraint — soft, learned sparsity helps a little, hard selection hurts, and the dominant limitation remains spatial transfer.

4.5 Per-class performance and persistent confusions

Figure 15 breaks out per-class holdout F1 for the best transferer (TabNet), a robust tree model (XGBoost field), a robust linear baseline (logistic regression), and the largest faller (CNN-BiLSTM). The gap is highly class-dependent. All models classify the majority crop lucerne/medics and the spectrally distinct canola well (TabNet F1 0.88 and 0.81 respectively), but accuracy on the minority cereals is poor and is where the transfer loss concentrates: TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small grain grazing. CNN-BiLSTM underperforms across *all* classes, not just the rare ones, confirming a generalization rather than an imbalance failure. The wheat–barley and small-grain-grazing confusions are consistent with their spectral overlap (Figure 4) and persist across every model and paradigm.

5 Discussion

Our results demonstrate that, in this setting, the conclusion one draws about which model is “best” depends almost entirely on whether evaluation is in-region or spatially out-of-sample — and that a model’s transfer behaviour is predictable from its inductive bias.

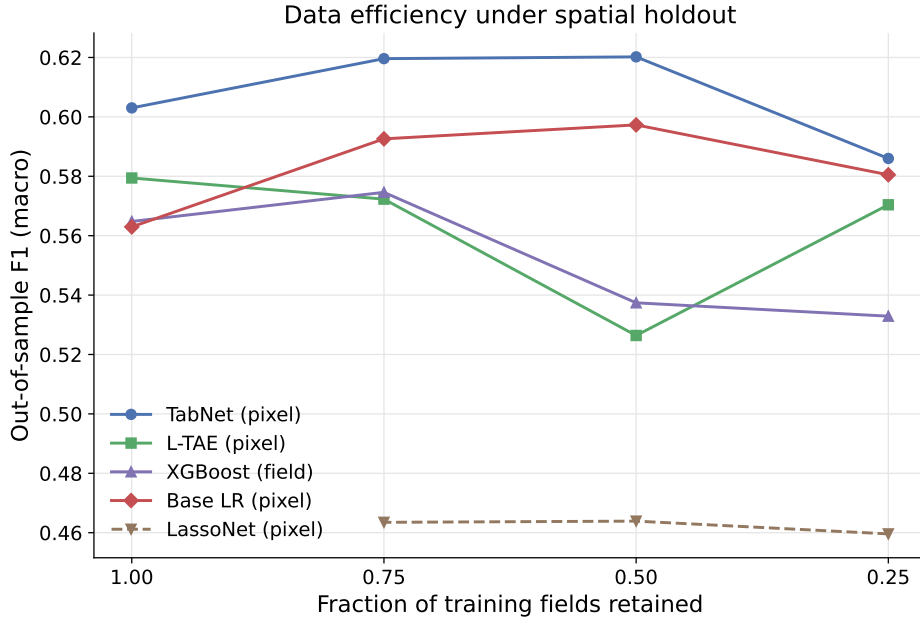


Fig. 14: Out-of-sample macro-F1 as the number of training fields is reduced to 75%, 50%, and 25%. Robust models (TabNet, logistic regression) are nearly flat; LassoNet is weak throughout.

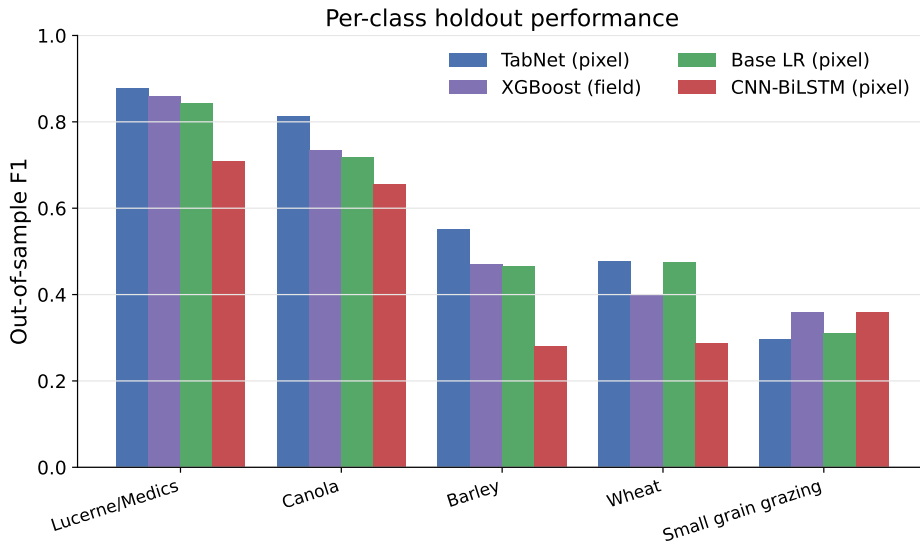


Fig. 15: Per-class out-of-sample F1 for four representative models. Lucerne/medics and canola are classified well by all; barley, wheat, and small grain grazing concentrate the errors. CNN-BiLSTM degrades across every class.

Why tree-like inductive bias transfers.

The single clearest pattern in our results is that transfer tracks *inductive bias*, not model family or capacity per se. Figure 16 groups every model by inductive bias and plots its generalization gap. The ordering is stark: linear models lose the least (mean $\Delta = -0.05$), tree-based models little (mean -0.11), while dense temporal/patch deep nets lose the most (mean -0.22) and SMOTE-augmented stacking is similarly fragile (-0.21). Random Forest and gradient-boosted trees have been the remote-sensing workhorse for two decades precisely because their sparse, axis-aligned splits select a few informative features at each node and ignore the rest, which resists overfitting to the idiosyncratic, region-specific structure of any one scene [15, 28]. Our results show that this same property is what governs *spatial* transfer. Crucially, the best out-of-sample model, TabNet, is a neural network built explicitly to emulate this behaviour: its Attentive Transformer emits sparse (sparsemax) feature masks that perform instance-wise, axis-aligned feature selection at each decision step, mirroring tree splits while remaining differentiable [33]. TabNet thus clusters with the trees, not with the dense sequence models, and inherits their transferability. By contrast, CNN-BiLSTM, TempCNN, and L-TAE build dense distributed representations over the full temporal sequence with no built-in feature sparsity; they fit the training region’s phenology and atmosphere tightly and transfer poorly.

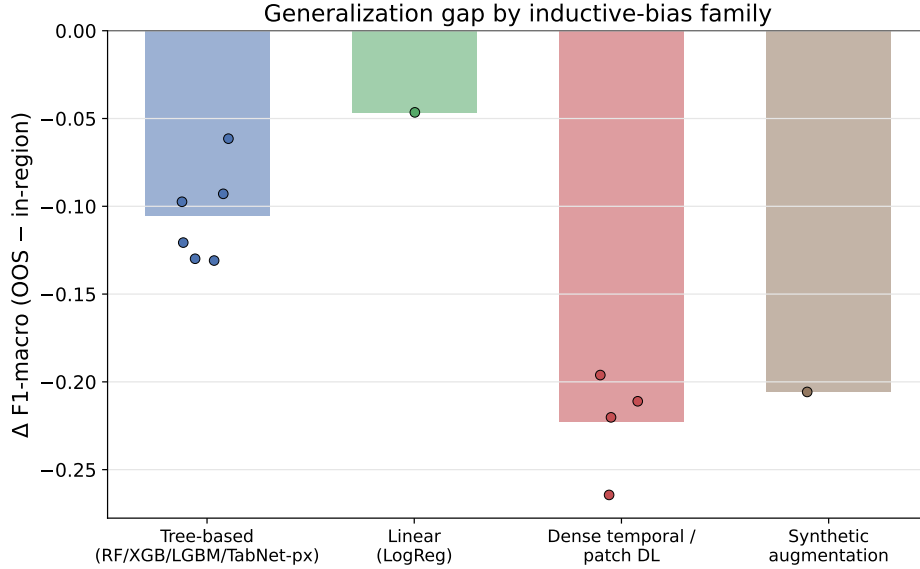


Fig. 16: Generalization gap (Δ macro-F1, out-of-sample minus in-region) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch deep nets and synthetic augmentation overfit the training region. Bars show family means; points are individual models.

Two robustness levers.

We are careful not to overclaim a single tidy axis, because two caveats sharpen rather than weaken the story. First, it is *pixel-level* TabNet that wins: the field-aggregated TabNet variants are among the worst transferers (Table 3), because aggregation removes the per-pixel sample volume that the sparse model exploits. Second, L-TAE — an attention model — transfers well *when* applied at the field level ($\Delta = -0.07$). These observations point to two distinct, partly substitutable routes to robustness. The first is a **sparse, tree-like inductive bias** baked into the model (trees, TabNet), which confers transfer intrinsically. The second is **variance reduction imposed from outside** via field-level aggregation (Section 4.2), which averages away region-specific pixel noise and rescues otherwise-fragile dense models. A model that already possesses the first lever gains nothing — and may lose sample volume — from the second, which is exactly why TabNet prefers pixels and L-TAE prefers fields. The fragile cluster (CNN-BiLSTM, 3D CNN, SMOTE stacking) combines high capacity, dense temporal representation, and (for SMOTE) synthetic interpolation, with neither lever engaged.

Automated feature extraction and computational cost.

The classical pipelines paired with `xr.fresh` were not only robust but cheap: the field-level Voting and Stacking ensembles transfer as well as the best deep models (OOS $F1_m$ 0.55–0.56) while training in under two minutes, versus hours for the temporal deep nets (Table 3). To our knowledge this is the first evaluation of `xr.fresh` in a crop-classification context, and the result is encouraging: automated time-series feature extraction yields a representation that, fed to a tree ensemble, both matches deep models on transfer and inherits the trees’ robustness — a favourable combination for operational, resource-constrained deployment.

Patch-level spatial modeling.

Patch-level 3D CNN models did not exceed pixel-level models in-region and transferred poorly ($F1_m = 0.48$, $\Delta = -0.22$). This is partly structural: at Sentinel-2’s 10–20 m resolution, fields in the Western Cape (median ~ 7 –11 ha) are often small enough that a fixed patch grid covers multiple fields or mixed-boundary pixels, introducing label noise, while the dense convolutional representation again offers no transfer-protective sparsity.

Persistent difficulty with small grain grazing and barley.

Small grain grazing, barley, and wheat are the hardest classes for every model and paradigm (Figure 15), and they concentrate the out-of-sample loss. Small grain grazing has both the fewest distinctive observations and a spectral-temporal signature that closely resembles the other cereals, particularly early in the season; wheat–barley confusion is similarly persistent and consistent with their spectral overlap (Figure 4). That even the robust models struggle here — and that CNN-BiLSTM degrades across *all* classes, not only the rare ones — indicates the limitation is partly intrinsic class similarity and partly spatial transfer, not class imbalance alone. Targeted collection for these classes, and richer temporal sampling, are the most promising remedies.

Limitations.

Several limitations constrain these results. First, the study uses a single growing season (2017) and a single holdout tile; while this already exposes a large spatial gap, multi-region and multi-year holdouts would test whether the inductive-bias ordering we report is stable across more diverse shifts, and we frame our finding as a method-level conclusion about transfer rather than a claim about absolute achievable accuracy on this small benchmark. Second, the labels derive from a public Radiant Earth MLHub dataset rather than our own field campaign; this aids reproducibility but means we inherit its digitizing and survey conventions. Third, we use Sentinel-2 optical data only. The exclusion of SAR is a deliberate scoping choice — to isolate the optical time-series signal and maximize applicability where only optical data exist — but it limits the achievable ceiling, particularly for optically similar cereals; SAR fusion is a clear next step (below). Finally, EVI was computed from monthly median composites rather than simultaneous acquisitions (Section 4); our feature-importance analysis indicates the models rely far more on raw bands than on EVI, so this does not drive the conclusions, but a calibrated index would require per-date gap-filling.

6 Conclusion

This study evaluated nineteen classical, deep learning, and patch-based crop classifiers in the Western Cape of South Africa under two protocols: conventional field-wise cross-validation within the training region, and a spatially disjoint holdout tile. The central finding is that these protocols disagree sharply about which model is best. Under in-region validation the dense temporal deep nets lead, and conventional field-level evaluation singles out a CNN-BiLSTM ensemble as best (macro-F1 0.72); under spatial transfer CNN-BiLSTM suffers the largest degradation of any model (macro-F1 0.46) and is among the worst, while a TabNet ensemble becomes the best single model (macro-F1 0.60, Kappa 0.54) and simple classical baselines — gradient-boosted trees and logistic regression — transfer with the smallest gaps.

We argue this reordering is mechanistic rather than incidental: the sparse, axis-aligned feature selection that made tree ensembles the remote-sensing default is precisely the inductive bias that confers spatial transfer, and TabNet inherits it through attentive sparsemax feature masks while dense temporal deep nets do not. Field-level aggregation provides a second, substitutable robustness lever — variance reduction that rescues dense models but is redundant for sparse ones. The most transferable models are also the most data-efficient, retaining near-full holdout accuracy on a quarter of the training fields.

For the remote-sensing community the practical message is concrete: report spatial-holdout performance, not just within-region cross-validation, because the latter is optimistic and can invert model rankings; prefer parsimonious, feature-selecting models (tree ensembles, TabNet) when the goal is mapping unseen terrain; and use field-level aggregation to harden dense models when they are required. Across all models, small-grain-grazing and wheat-barley confusion remain the dominant residual errors, calling for richer temporal sampling and targeted collection for these classes. The most productive future directions are SAR-optical fusion to raise the achievable

ceiling for spectrally similar cereals, multi-region and multi-year holdouts to test the stability of the inductive-bias ordering, and domain-adaptation methods that explicitly optimize for transfer. Higher-cadence optical platforms such as PlanetScope could further mitigate the cloud-cover and temporal-resolution limits encountered here.

Conflict of Interest

The authors declare that they have no competing interests.

Data Availability

This study relied on multiple sources of data. Ground-truth crop type labels were obtained from Radiant Earth’s MLHub platform [3], while multi-temporal Sentinel-2 imagery was extracted using the Google Earth Engine API. Since the raw data was processed and combined in various ways specific to this study, the resulting dataset used for model training and evaluation is not hosted publicly but can be shared by the authors upon reasonable request for academic or research purposes.

Code Availability

All source code used for data processing, modeling, and evaluation is publicly available [4].

A Baseline classical model definitions

For completeness we reproduce the standard formulations of the four baseline classifiers described in Section 3.1. For a feature vector $\mathbf{x}_i$ and class $c \in \{1, \dots, C\}$:

Logistic Regression [23] (multinomial / softmax):

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{x}_i + b_c)}{\sum_{c'=1}^C \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'})},$$

with per-class weight vectors $\mathbf{w}_c$ and biases b_c .

Random Forest [24]: majority vote over K decision trees,

$$\hat{y}_i = \text{mode} \{T_1(\mathbf{x}_i), \dots, T_K(\mathbf{x}_i)\}.$$

LightGBM [25]: additive gradient boosting with learning rate η and tree f_t ,

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(\mathbf{x}_i).$$

XGBoost [26, 27]: regularized boosting objective,

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f_k) = \gamma T_k + \frac{1}{2} \lambda \sum_j w_j^2,$$

where l is the per-sample loss, T_k the number of leaves in tree f_k , w_j the leaf weights, and γ, λ regularization coefficients.

B Performance metric definitions

Let N be the number of samples, C the number of classes, and n_c the support of class c . **Cohen’s Kappa**:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_o = \frac{\#\text{correct}}{N}, \quad p_e = \sum_{c=1}^C \frac{n_{c,\text{true}}}{N} \cdot \frac{n_{c,\text{pred}}}{N}.$$

Macro-averaged F1 (primary metric) and **weighted F1**:

$$\text{F1}_m = \frac{1}{C} \sum_{c=1}^C \text{F1}_c, \quad \text{F1}_{\text{weighted}} = \sum_{c=1}^C \frac{n_c}{N} \text{F1}_c, \quad \text{F1}_c = \frac{2 \text{Prec}_c \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}.$$

Log-loss (cross-entropy), with one-hot labels $y_{i,c}$ and predicted probabilities $p_{i,c}$:

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \ln(p_{i,c}).$$

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